Let the mind manage the body
Que l'esprit gère le corps

Index Number: ....

NATIONAL CERTIFICATE OF EDUCATION

2024

CHEMISTRY (N530)

TIME: 45 MINUTES

Candidates answer on the Question Paper.
Additional Material: Ruler, Calculator

READ THESE INSTRUCTIONS FIRST

1. Write your index number in the space provided above.
2. Write in dark blue or black ink. Do not use correction fluid.
3. You may use a soft pencil for any diagram, graph or rough working.
4. Diagrams are not drawn to scale unless otherwise specified.
5. Any rough working should be done in this booklet.
6. Answer **ALL** questions.
7. This document consists of **5** questions printed on **9** pages, numbered **2** to **10**.
8. A copy of the Periodic Table is provided on page **11.**
9. The total number of marks for this paper is **50**.

For Examiners' use							
Question No.	**1**	**2**	**3**	**4**	**5**	**Total**	**Signature**
Examiner							
Team Leader							
CE/ACE							

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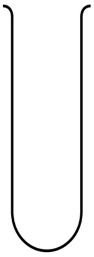
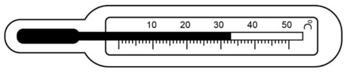
Marks

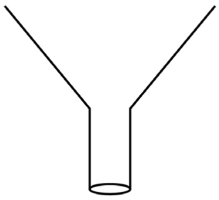
Question 1 (10 marks)

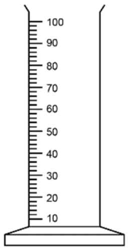
Circle the correct answer. Each item carries one mark.

1. Which one of the following is used to measure **temperature**?

A 

B 

C 

D 

2. Fig. 1.1 shows a molecule of ammonia.

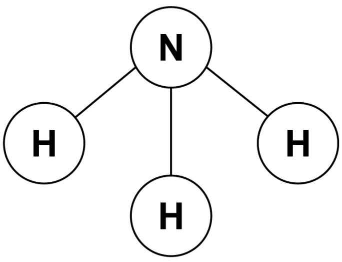


Fig. 1.1

How many **hydrogen** atoms are there in a molecule of ammonia?

A 1

B 2

C 3

D 4

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Marks

3. What is the valency of aluminium?

A 1 **B** 2

C 3 **D** 4

4. Fig. 1.2 shows a methane molecule.

What is the formula of the methane molecule?

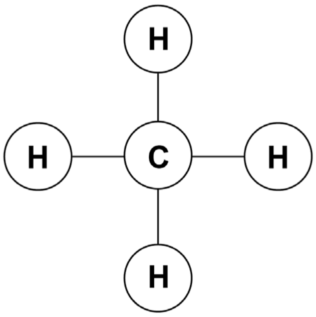

Fig. 1.2

A C_2H **B** C_4H

C CH_2 **D** CH_4

5. The ozone layer is depleted by ………………….. .

A chlorofluorocarbons

B carbon monoxide

C oxides of nitrogen

D sulfur dioxide

6. Which of the following will cause eutrophication?

A Smoke from factories

B Fertilisers from agriculture

C Oil spillage from ships

D Plastic objects in rivers

Marks

7. Fig. 1.3 shows a piece of zinc in dilute hydrochloric acid.

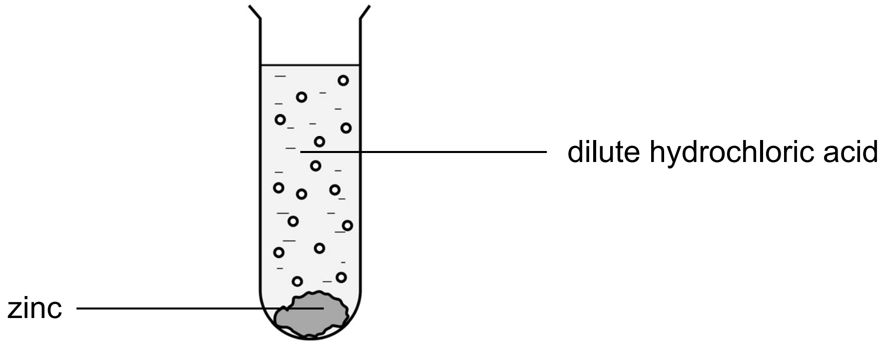

Fig. 1.3

Which gas is produced when zinc reacts with dilute hydrochloric acid?

A Oxygen

B Sulfur dioxide

C Hydrogen

D Carbon dioxide

8. Which one of the following metals **does not** burn or glow when heated?

A Copper

B Iron

C Magnesium

D Zinc

9. Which substance is used to control the acidity of soil?

A Calcium hydroxide

B Ammonium sulfate

C Potassium nitrate

D Nitric acid

10. Which one of the following salts is **insoluble** in water?

A Ammonium hydroxide

B Calcium nitrate

C Zinc sulfate

D Magnesium carbonate

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Question 2 (7 marks)

Three separation techniques are given below:

Crystallisation **Sublimation** **Distillation**

(a) From the above list, choose the separation technique used to obtain:

(i) water from sea water …………………………………

(ii) iodine from a mixture of iodine and sand ………………………………… [2]

(b) The set-up for sublimation is shown in Fig. 2.1.

(i) Label the diagram by filling the empty boxes using the words given below.

Evaporating dish	**Bunsen burner**	**tripod**	**sublimate**	**beaker**

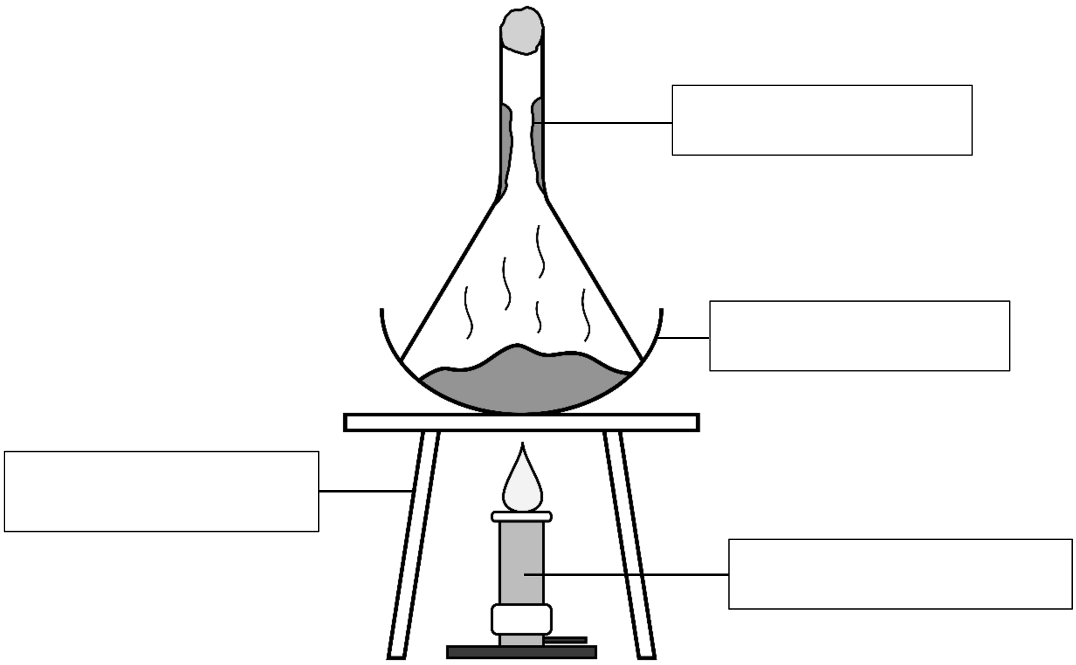

Fig. 2.1

[4]

(ii) What precaution needs to be taken during sublimation to prevent vapours from escaping?

…………………………………………………………………………………………….. [1]

E
TL
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Marks

Question 3 (10 marks)

(a) (i) Complete the table below.

	Potassium	Oxygen
Symbol	………………..	………………..
Valency	………………..	………………..

[4]

(ii) Hence, work out the formula of potassium oxide.

………………………………… [1]

(b) Work out the formula of each of the following compounds.

(i) Calcium chloride

………………………………… [1]

(ii) Iron (III) sulfate

………………………………….. [1]

(c) Complete the following word equations.

(i) Aluminium + Chlorine $\longrightarrow$ ……………………….………………………

(ii) Magnesium + Oxygen $\longrightarrow$ ………………………….…………………

(iii) Hydrogen + Oxygen $\longrightarrow$ ……………………………….……………

[3]

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Question 4 (11 marks)

Table 1 below shows the amount of carbon dioxide in the atmosphere from the year 1800 to the year 2000.

Table 1

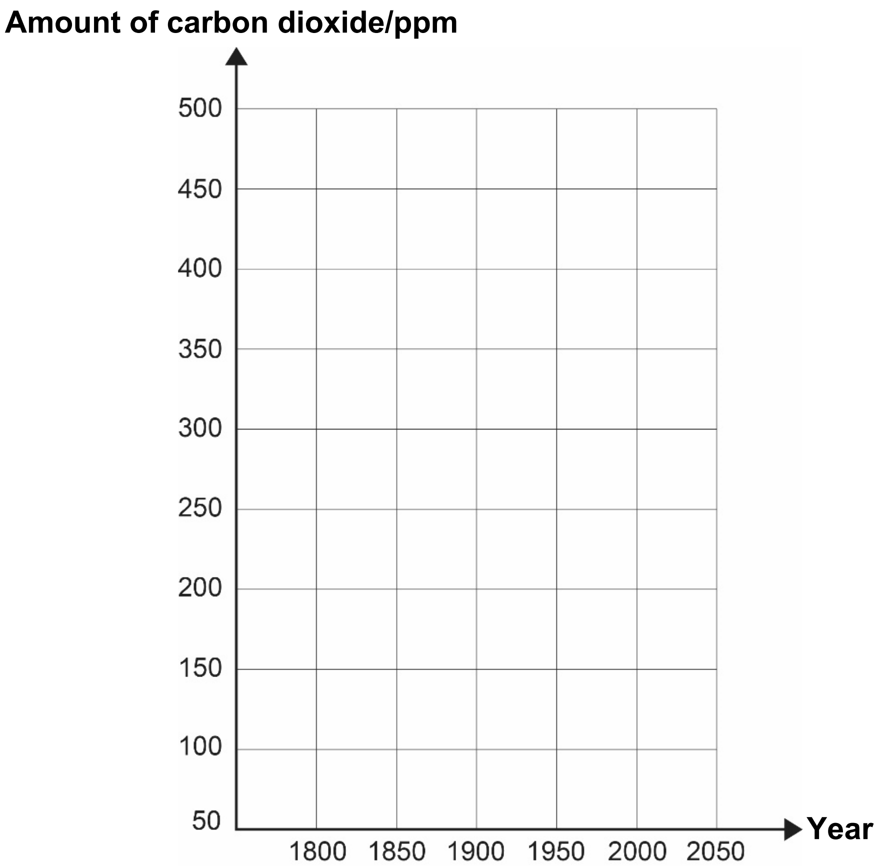
Year	1800	1850	1900	1950	2000
Amount of carbon dioxide/ppm	100	150	200	230	450

(a) Calculate the increase in the amount of carbon dioxide from the year 1950 to the year 2000.

……………………………………… [1]

(b) Using the information in **Table 1**,

(i) plot a graph of the amount of carbon dioxide against year on the graph given below. [2]



(ii) Using the graph, describe the trend in the amount of carbon dioxide from the year 1800 to the year 2000.

…………………………………………………………………………………………….

……………………………………………………………………………………………. [2]

Marks

(c) Carbon dioxide is a greenhouse gas.

(i) What is a greenhouse gas?

…………………………………………………………………………………………….. [1]

(ii) Name **another** greenhouse gas.

…………………………………… [1]

(d) Give two activities that have contributed to an increase in the amount of carbon dioxide in the atmosphere.

1. ………………………………………………………………………………………….

2. ………………………………………………………………………………………….

[2]

(e) Give two ways in which an increase in the amount of carbon dioxide impacts the environment negatively.

1. ………………………………………………………………………………………….

………………………………………………………………………………………….

2. ………………………………………………………………………………………….

………………………………………………………………………………………….

[2]

E
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CE

Question 5 (12 marks)

(a) Sodium reacts with water to form sodium hydroxide and a gas.

(i) Name the gas produced.

………………………………………………………………………………………… [1]

(ii) Give two observations when sodium reacts with water.

1. ……………………………………………………………………………...

2. ……………………………………………………………………………... [2]

(iii) Write down a balanced chemical equation for the above reaction.

………………………………………………………………………………………… [2]

(iv) What is the name given to the reaction between sodium hydroxide and an acid?

………………………………………………………………………………………… [1]

(b) The table below provides information about the reactivity of metal **X** and metal **Y** with cold water and steam.

Metal	Reaction with	
	cold water	**steam**
X	No reaction	reacts
Y	No reaction	No reaction

Using the above information, name

Metal **X**: ……………………………………

Metal **Y**: ……………………………………

[2]

Marks

(c) Complete **Table 2**.

The first one has been done for you.

Table 2

Name of salt	**Use of salt**	**Solubility of salt**
Sodium chloride	For enhancement of taste	**Soluble**
………………..………………	Baking	…………..……………
Calcium sulfate	………………….………..	…………………………..

[4]

E

TL

CE

The Periodic Table of Elements

Group																	
I	II											III	IV	V	VI	VII	VIII
							H hydrogen										He helium
Li lithium	Be beryllium											B boron	C carbon	N nitrogen	O oxygen	F fluorine	Ne neon
Na sodium	Mg magnesium											A*l* aluminium	Si silicon	P phosphorus	S sulfur	C*l* chlorine	Ar argon
K potassium	Ca calcium	Sc scandium	Ti titanium	V vanadium	Cr chromium	Mn manganese	Fe iron	Co cobalt	Ni nickel	Cu copper	Zn zinc	Ga gallium	Ge germanium	As arsenic	Se selenium	Br bromine	Kr krypton
Rb rubidium	Sr strontium	Y yttrium	Zr zirconium	Nb niobium	Mo molybdenum	Tc technetium	Ru ruthenium	Rh rhodium	Pd palladium	Ag silver	Cd cadmium	In indium	Sn tin	Sb antimony	Te tellurium	I iodine	Xe xenon
Cs caesium	Ba barium	lanthanoids	Hf hafnium	Ta tantalum	W tungsten	Re rhenium	Os osmium	Ir iridium	Pt platinum	Au gold	Hg mercury	T*l* thallium	Pb lead	Bi bismuth	Po polonium	At astatine	Rn radon
Fr francium	Ra radium	actinoids	Rf rutherfordium	Db dubnium	Sg seaborgium	Bh bohrium	Hs hassium	Mt meitnerium	Ds darmstadtium	Rg roentgenium	Cn copernicium		F*l* flerovium		Lv livermorium		

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